

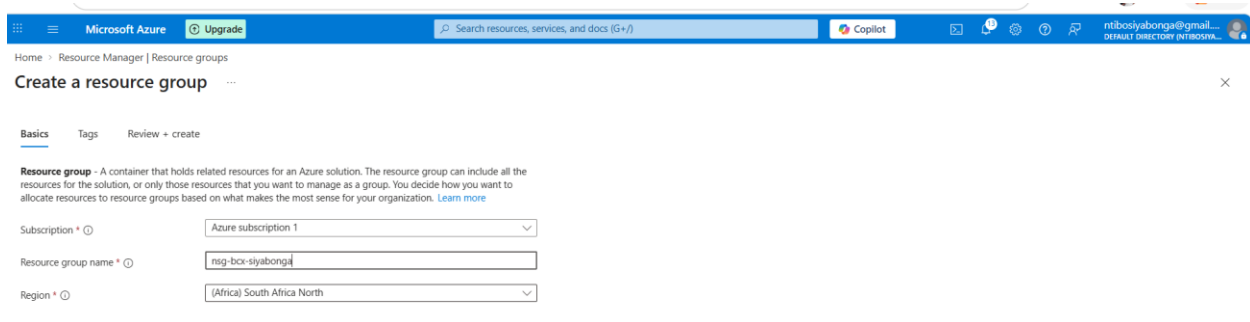
TASK 3: SECURITY IMPLEMENTATION

Chapter 08

Step 1 – Explain Shared Responsibility

In Azure, Microsoft is responsible for the **security of the cloud**, including physical infrastructure, host OS, and network. BCX, as the administrator, is responsible for **security in the cloud**, including configuring identity and access, secrets, policies, network security, and resource settings. Even in a managed cloud, misconfigured users or resources can introduce risks, so BCX must actively manage access and governance.

Step 2 – Configure Network Security Group (NSG)



Microsoft Azure Upgrade Search resources, services, and docs (G+/I) Copilot nsiibosiyabonga@gmail... DEFAULT DIRECTORY INTERNAL...

Home > Resource Manager | Resource groups

Create a resource group ...

Basics Tags Review + create

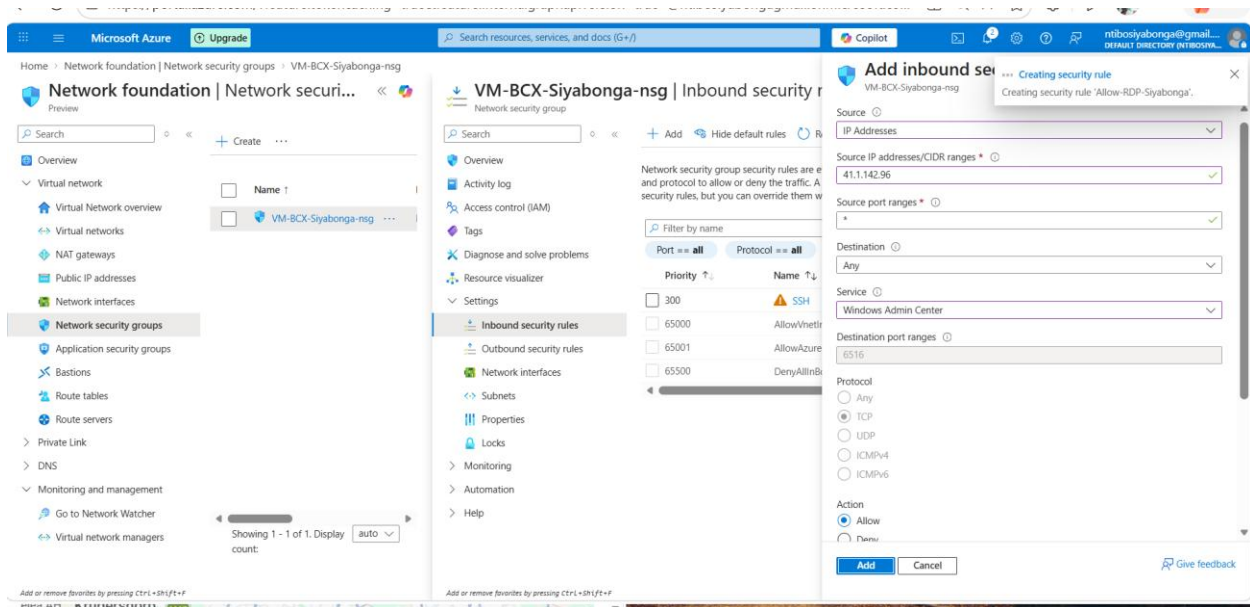
Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * Azure subscription 1

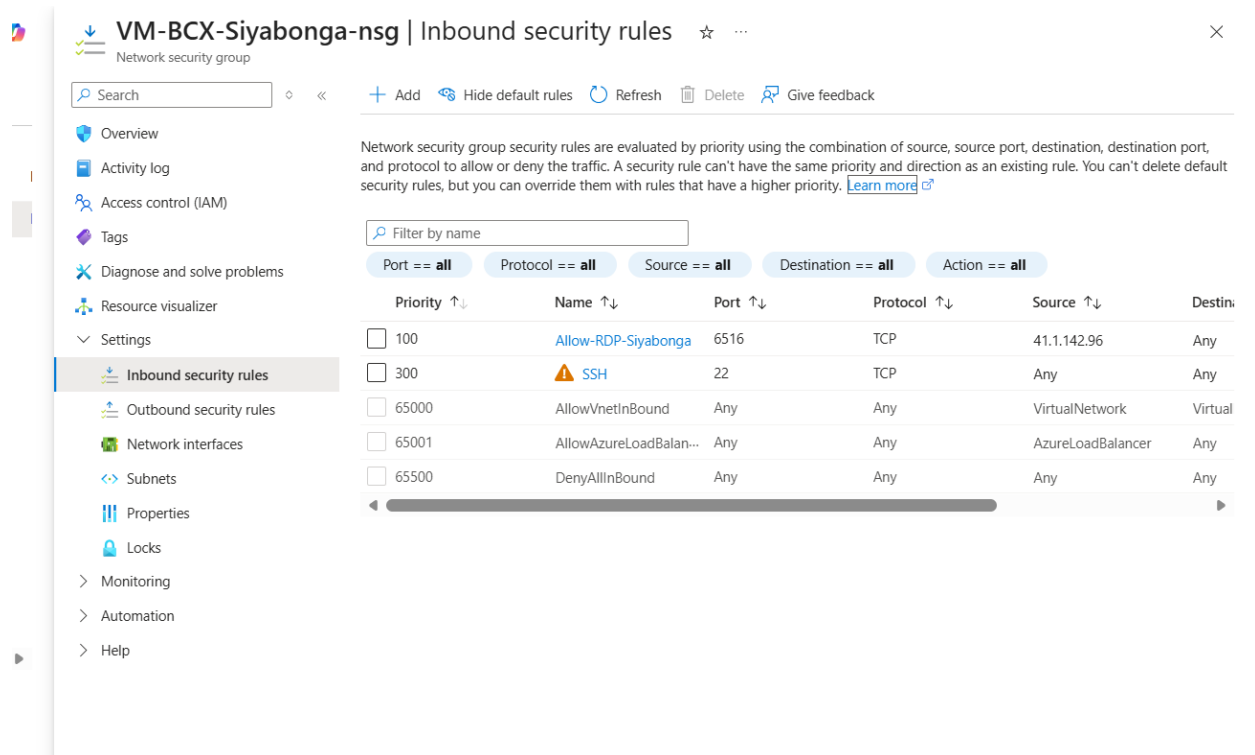
Resource group name * nsg-bcx-siyabonga

Region * (Africa) South Africa North

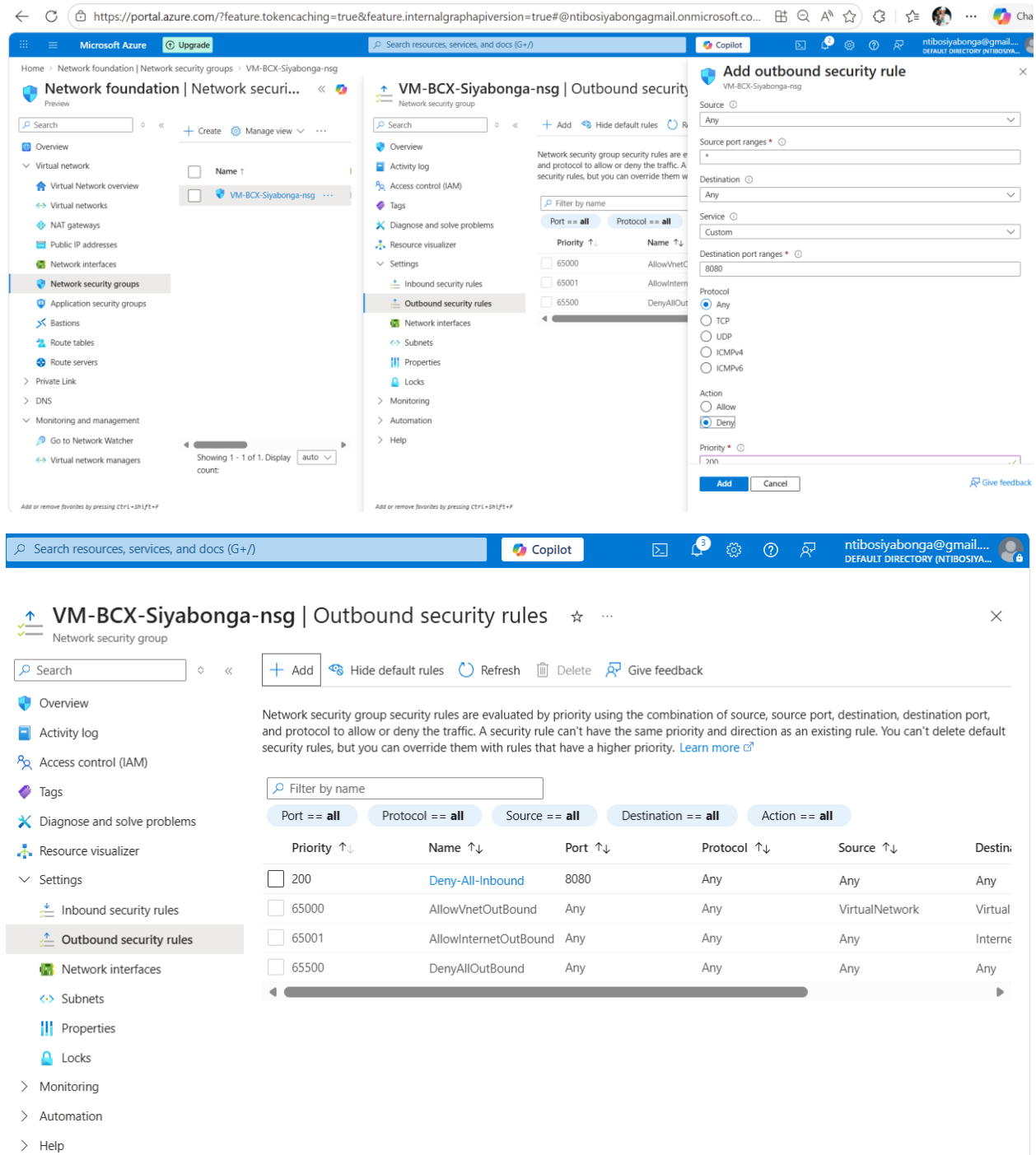
Rule 1: Allow SSH from your IP only



...&feature.internalgraphapiversion=true#@ntibosiyabongagmail.onmicrosoft.co...



Rule 2: Deny all other inbound traffic



The screenshot displays the Microsoft Azure portal interface for configuring a Network Security Group (NSG) rule. The main pane shows the 'Add outbound security rule' dialog for the NSG 'VM-BCX-Siyabonga-nsg'. The rule is named 'Deny-All-Inbound', has a priority of 200, and is configured to deny all traffic from any source to any destination on any port using any protocol. The left sidebar shows the 'Network security groups' section, and the right sidebar shows the 'Outbound security rules' list.

Network security group: VM-BCX-Siyabonga-nsg | Outbound security rules

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. [Learn more](#)

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

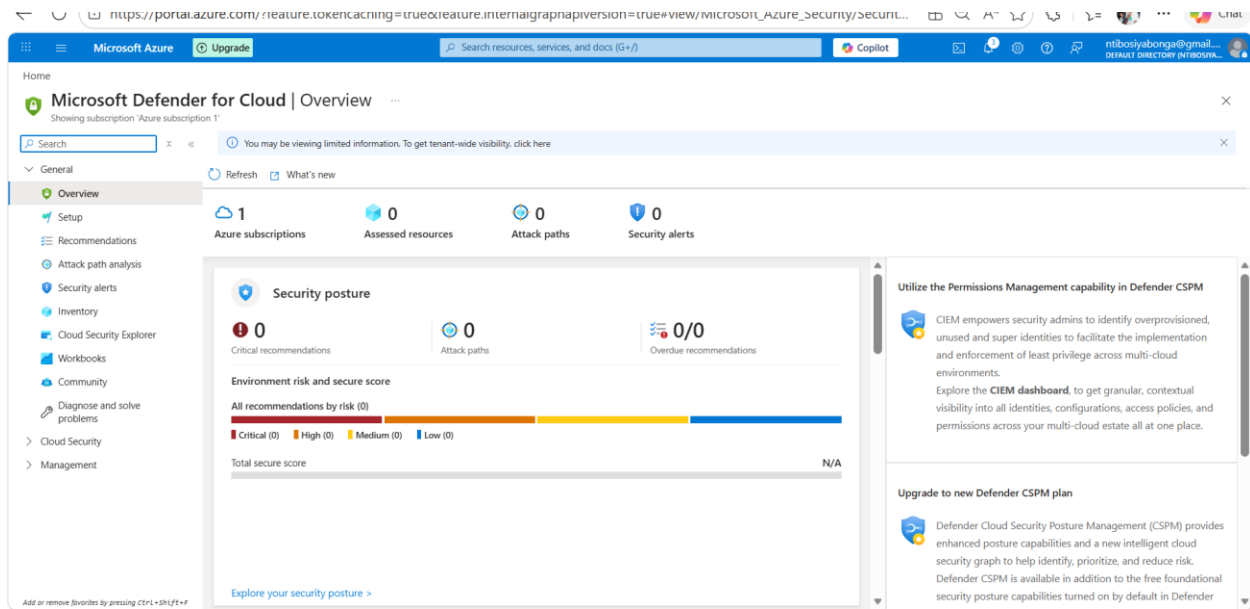
Priority	Name	Port	Protocol	Source	Destination
200	Deny-All-Inbound	8080	Any	Any	Any
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	Virtual
65001	AllowInternetOutBound	Any	Any	Any	Interne
65500	DenyAllOutBound	Any	Any	Any	Any

NSGs control inbound and outbound traffic to resources, acting as a virtual firewall. Rule-based filtering ensures only authorized access is allowed.

Step 3 – Azure Firewall Discussion

Azure Firewall is a cloud-native firewall that protects network traffic at the application and network layers. Unlike NSGs, which are rule-based for subnets or NICs, Azure Firewall can enforce fully stateful, centralized filtering, NAT rules, and application-level rules for enterprise networks.

Step 4 – Microsoft Defender for Cloud



The screenshot shows the Microsoft Defender for Cloud Overview page. The top navigation bar includes the Microsoft Azure logo, an Upgrade button, a search bar, and a Copilot icon. The main content area displays the following metrics:

- Azure subscriptions: 1
- Assessed resources: 0
- Attack paths: 0
- Security alerts: 0

The Security posture section shows:

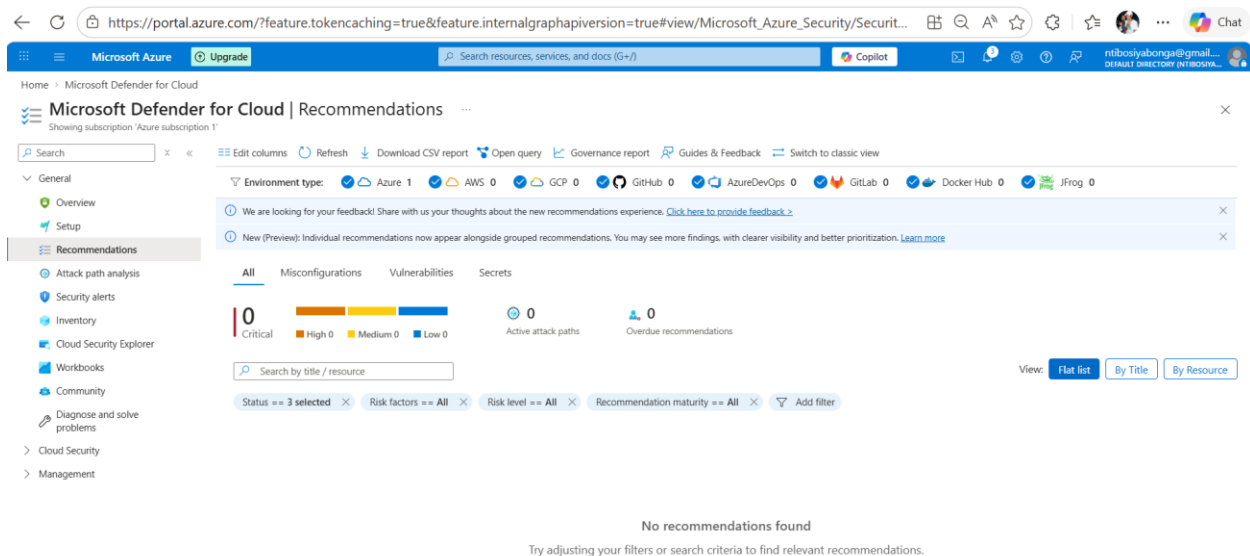
- Critical recommendations: 0
- Attack paths: 0
- Overdue recommendations: 0/0

The Environment risk and secure score section shows:

- All recommendations by risk (0): Critical (0), High (0), Medium (0), Low (0)
- Total secure score: N/A

The right sidebar contains two sections:

- Utilize the Permissions Management capability in Defender CSPM**: A message about CIEM (Cloud Infrastructure Entitlement Management) empowering security admins to identify overprovisioned, unused, and super identities.
- Upgrade to new Defender CSPM plan**: A message about Defender Cloud Security Posture Management (CSPM) providing enhanced posture capabilities and a new intelligent cloud security graph.



The screenshot shows the Microsoft Defender for Cloud Recommendations page. The top navigation bar is the same as the Overview page. The main content area displays the following metrics:

- Environment type: Azure 1, AWS 0, GCP 0, GitHub 0, AzureDevOps 0, GitLab 0, Docker Hub 0, Jfrog 0
- All recommendations: 0
- Active attack paths: 0
- Overdue recommendations: 0

The Recommendations section shows:

- All recommendations: 0
- Active attack paths: 0
- Overdue recommendations: 0

The bottom of the page displays a message: "No recommendations found. Try adjusting your filters or search criteria to find relevant recommendations."

Step 5 – Create Azure Key Vault

The image consists of two screenshots from the Azure Portal. The top screenshot shows the 'Create a key vault' wizard. The 'Subscription' is 'Azure subscription 1' and the 'Resource group' is 'RG-BCX-Siyabonga'. Under 'Instance details', the 'Key vault name' is 'kv-bcx-siyabonga', the 'Region' is 'South Africa North', and the 'Pricing tier' is 'Standard'. The 'Recovery options' section shows 'Soft delete' is 'Enabled' with 'Days to retain deleted vaults' set to '90'. Navigation buttons at the bottom are 'Previous', 'Next', and 'Review + create'. The bottom screenshot shows the 'Microsoft Defender for Cloud | Security alerts' dashboard. It displays a summary of alerts: 0 Open alerts, 0 Active alerts, 0 In progress alerts, and 0 Affected resources. A table with columns for Severity, Alert name, Affected resource, Resource Group, Activity start time, Last updated time, MITRE ATT&CK tactics, and Status is shown, but it contains no data. The left sidebar lists navigation options like Overview, Setup, Recommendations, and Security alerts.

Defender for Cloud monitors resources across Azure, hybrid, and multi-cloud environments. It provides security recommendations, threat alerts, and an overall secure posture for enterprise workloads.

← ↻ <https://portal.azure.com/?feature.tokencaching=true&feature.internalgraphapiversion=true#view/HubsExtension/DeploymentDetail...> Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot Chat

Home kv-bcx-siyabonga | Overview

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : kv-bcx-siyabonga

Subscription : Azure subscription 1

Resource group : RG-BCX-Siyabonga

Start time : 4/8/2026, 3:34:15 PM

Correlation ID : cfd2877-a2dd-4146-b28e-8ac8a3549f18

Deployment details

Resource	Type	Status	Operation details
kv-bcx-siyabonga	Key vault	OK	Operation details

Next steps

[Go to resource](#)

Cost management

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Microsoft Defender for Cloud

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Home > kv-bcx-siyabonga | Secrets

Create a secret

Upload options Manual

Name * AdminPassword

Secret value * *

Content type (optional)

Set activation date

Set expiration date

Enabled Yes No

Tags 0 tags

Create Cancel

BCX should store sensitive information such as admin passwords, API keys, and database connection strings in Azure Key Vault rather than hardcoding them into scripts or applications. This ensures secrets are centrally managed, encrypted, and access-controlled

Step 6 – Defense in Depth Mapping

Layer	BCX Example Control
Physical	Azure-managed data centers
Identity & Access	MFA, RBAC, Conditional Access
Perimeter	Azure Firewall
Network	NSG rules, VPN Gateway
Compute	VM hardening, patching
Application	Secure coding, App Service security
Data	Key Vault, encryption at rest & in transit

Step 7 – Zero Trust Application

Zero Trust assumes no user or device is inherently trusted, even inside the network. BCX applies explicit verification, least privilege access, assume breach mentality, and continuous authentication to ensure resources are protected.

Step 8 – SIEM and Threat Monitoring (Azure Sentinel)

Azure Sentinel is a cloud-native SIEM solution that collects data from Azure, on-premises, and other clouds. It aggregates, analyzes, and detects threats. BCX can use Sentinel to monitor multiple client environments, respond to incidents, and gain insights into potential attacks.